

CLIMATOLOGY

Century-long West Antarctic snow accumulation changes induced by tropical teleconnections

Kai Man^{1,2}, Jürg Luterbacher³, David M. Holland⁴, Naiming Yuan^{5,6,7}, Lei Geng⁸, Yetang Wang⁹, Yonggang Liu¹⁰, Guitao Shi^{11,12}, Yurong Hou^{1,2}, Wenju Cai^{13,14,15,16*}, Xichen Li^{1*}

Ice core measurements reveal dipole-like snow accumulation trends over West Antarctica throughout the 20th century, with an increase of >2000 billion metric tons over the Antarctic Peninsula and Ellsworth Land but a decrease of ~500 billion metric tons over Marie Byrd Land. Although atmospheric teleconnections were frequently revealed, linking variability between tropics and higher latitudes on interannual and decadal timescales, centennial-scale teleconnection is absent from literature. Here, using statistical analysis and numerical experiments, we reveal that changes of tropical oceans throughout the 20th century drive the long-term Antarctic snowfall trend. A pronounced warming over the tropical Atlantic and a moderate cooling over the equatorial Pacific have driven an adjustment of moisture transport and thus snowfall pattern in West Antarctica. Our study reveals a centennial tropical-polar teleconnection, producing long-term trends with opposing changes across the regions. Remote forcing from the tropics increased the mass accumulation over Antarctica, balanced rapid iceshelf thinning in recent decades, contributing to global sea-level changes.

INTRODUCTION

Since 1990s, the mass loss of the Antarctic ice sheet has accelerated markedly (1, 2), as evidenced by in situ and satellite observations, releasing more than 2000 billion metric tons (Gt) of freshwater (2) and contributing to global sea level rise (3). This mass loss is primarily driven by the accelerated glacier flow in coastal regions, caused by the intense melting of ice shelves (4, 5). Concurrently, the dynamic mass loss of the Antarctic ice sheet was balanced by snowfall over the Antarctic continent. In the 20th century, the surface mass balance over the Antarctic Ice Sheet exhibited an increasing trend due to increased snowfall (6, 7). To some extent, the increased snowfall has mitigated the dynamic mass loss of the Antarctic Ice Sheet and has limited sea level rise (6, 7).

However, the changes in surface mass balance of the Antarctic Ice Sheet during the 20th century reveal a complex regional pattern (6–10). An increased snow accumulation occurred over the Antarctic Peninsula and Ellsworth Land (6–9); in contrast, East Antarctica

has been relatively stable (10). Regional disparities are also evident in broader climate changes in the Antarctic atmosphere, ocean, and cryosphere (11), with West Antarctica experiencing surface warming rates approximately twice the global average (12, 13), and most of the ice sheet mass loss localized to this region where subsurface warm water intrusion (1, 14) occurs. These regionally varied changes have profound implications for global climate, influencing global sea level (15, 16), ocean circulation (17, 18), energy balance (19), and carbon cycle (20).

Many regional characteristics of Antarctic climate change are influenced by remote forcing from the tropics through atmospheric teleconnections (21). Variability on interannual and decadal timescales over the tropical oceans may generate stationary Rossby wave trains (22, 23), propagating to the polar region and driving an adjustment of the large-scale atmospheric circulation, inducing anomalies in the Southern Annular Mode (SAM) and the Amundsen Sea Low (ASL). On interannual timescales, the El Niño–Southern Oscillation (ENSO) and Indian Ocean Dipole predominantly influence the Southern Hemisphere high-latitude surface air temperature (24) and sea ice (25, 26), while on decadal timescales, oceanic variability such as the Atlantic Multi-Decadal Oscillation (AMO) and the Interdecadal Pacific Oscillation (IPO) contribute to Antarctic climate change (27–30).

Tropical-polar interannual or decadal teleconnections operate, but whether these teleconnections work on centennial scales is unknown. This is largely because of the scarcity of observational data and the reduced reliability of reanalyses before the satellite era (~1979) (31). Furthermore, much of the scientific focus has been on the dominant role of the greenhouse gas-induced radiative forcing in driving long-term trends of the earth system (3). On centennial timescales, greenhouse forcing drives a global scale sea surface temperature (SST) warming pattern, which may obscure the centennial teleconnections between regional climates. The climate change patterns during the past century, especially the tropical oceans, show distinct regional features (32–36). For example, since the early 1900s, the tropical Atlantic and Indian Oceans have undergone continuous warming (34, 35), with average rates of about $0.066^{\circ} \pm 0.032^{\circ}\text{C/}$

¹Institute of Atmospheric Physics, Chinese Academy of Sciences, Beijing, China. ²University of Chinese Academy of Sciences, Beijing, China. ³Center for International Development and Environmental Research and Department of Geography, Justus Liebig University, Giessen, Germany. ⁴Courant Institute of Mathematical Sciences, New York University, New York, USA. ⁵School of Atmospheric Sciences, Sun Yat-sen University, Zhuhai, China. ⁶Key Laboratory of Tropical Atmosphere–Ocean System (Ministry of Education), Zhuhai, China. ⁷Southern Marine Science and Engineering Guangdong Laboratory, Zhuhai, China. ⁸School of Earth and Space Sciences, University of Science and Technology of China, Hefei, Anhui, China. ⁹College of Geography and Environment, Shandong Normal University, Jinan, China. ¹⁰Laboratory for Climate and Ocean–Atmosphere Studies, Department of Atmospheric and Oceanic Sciences, School of Physics, Peking University, Beijing, China. ¹¹State Key Laboratory of Estuarine and Coastal Research, East China Normal University, Shanghai, China. ¹²Key Laboratory of Geographic Information Science (Ministry of Education), School of Geographic Sciences, East China Normal University, Shanghai, China. ¹³Frontiers Science Center for Deep Ocean Multispheres and Earth System and Physical Oceanography Laboratory, Ocean University of China, Qingdao, China. ¹⁴Qingdao National Laboratory for Marine Science and Technology, Qingdao, China. ¹⁵State Key Laboratory of Marine Environmental Science & College of Ocean and Earth Sciences, Xiamen University, Xiamen, China. ¹⁶State Key Laboratory of Loess and Quaternary Geology, Institute of Earth Environment, Chinese Academy of Sciences, Xi'an, China.

*Corresponding author. Email: lixichen@mail.iap.ac.cn (X.L.); cwjresearch@gmail.com (W.C.)

decade and $0.064^{\circ} \pm 0.030^{\circ}\text{C}/\text{decade}$ (95% confidence interval), respectively. By contrast, the equatorial eastern Pacific has shown a slight cooling trend (32, 36).

Whether the century-long SST changes across the tropical oceans could physically link the changes over different areas of the Antarctic remains unknown. Using Antarctic ice core records and the historical ship-based SST observations over the tropical ocean, here, we find that the contrasting tropical SST trends drive the dipole-like snowfall trends over West Antarctica by modulating moisture advection in the lower troposphere.

RESULTS

A dipole-like trend pattern in West Antarctic snow accumulation

We calculate the Antarctic snow accumulation trends over the entire 20th century (1901–2000) using three state-of-the-art reconstruction datasets (6–8) (see the “Datasets” section in Materials and Methods) and ice-core records (table S1). We focus on the 20th century, because during that period, high-quality observational data are provided over both the tropical oceans (37) and Antarctica (10). Statistical analysis results (Fig. 1, A to E) show that during 1901–2000, a pronounced dipole-like pattern in West Antarctica as in dataset from (6) (Fig. 1, D and E) is characterized by a decrease in the accumulation rate in the west but an increase in the east. This pattern is corroborated by previous studies using ice core measurements (6, 7). East Antarctica, by contrast, shows smaller changes in the accumulation rate during the 20th century, with an area-weighted mean of -0.75 ± 2.8 mm/year per century (Fig. 1D). We validate this regional snow accumulation trend pattern using two other datasets (7, 8). The results show the same eastern-western contrast pattern, with a slightly larger amplitude (fig. S1). Below, we focus on the changes in the West Antarctic snow accumulation rate and their physical mechanisms.

Analysis of the reconstruction dataset from (6) (Fig. 1E) reveals that the accumulation rate over the Antarctic Peninsula and Ellsworth Land has increased at an average rate of 58.9 ± 26.3 mm/year per century. In contrast, the accumulation rate over Marie Byrd Land exhibited a statistically significant negative trend of -43.7 ± 25.3 mm/year per century. This east-west dipole pattern is similarly evident in the ice core records (triangles in Fig. 1E). For example, the Bruce Plateau ice core in the Antarctic Peninsula recorded a statistically significant and dramatic increase in accumulation rate of 842.7 ± 394.5 mm/year per century (Fig. 1C), whereas an ice core in Marie Byrd Land revealed a statistically significant decrease of -72.6 ± 41.6 mm/year per century (Fig. 1A). We also evaluate trends in snowfall and evaporation using two reanalysis datasets (see the “Datasets” section in Materials and Methods). The spatial distribution of the snowfall trends agrees well with that of the snow accumulation rate (fig. S2, A and B), while the evaporation trends (fig. S2, C and D) are relatively small and contribute less to the surface mass balance.

We illustrate the time series of integrated snow accumulation rate across western and eastern parts of West Antarctica (fig. S3). Results show a continuing increase in snow accumulation rate in the east (fig. S3C), while a decrease is observed in the west (fig. S3D) during the 20th century. The net effect is an overall increase in snow accumulation rate over West Antarctica (fig. S3A). This dipole-like trend pattern induces a disparity in the surface mass balance between the eastern and western parts of the West Antarctic Ice Sheet. To evaluate their contribution to mass changes in the West Antarctic

Ice Sheet, we calculate the total anomalous snow accumulation throughout the 20th century (Fig. 1, F to H). Until the end of 20th century, according to three datasets from (6), (7) and (8), 1943.1 to 2319.8 Gt of additional snow has accumulated over the Peninsula and Ellsworth Land (blue curve in Fig. 1 (F to H), see the “Cumulative mass change” section in Materials and Methods), in comparison to the mean accumulation rate between 1901 and 1920 (more results show in fig. S4 in comparison with the other periods). In contrast, the western part of West Antarctica, including Marie Byrd Land, experienced a mass loss of 472.5 to 938.1 Gt (red curve in Fig. 1, F to H). This resulted in a net increase of snow accumulation of 1234.5 to 1568.9 Gt (violet curve in Fig. 1, F to H) across West Antarctica. However, these datasets disagree with the sign of the East Antarctic Ice Sheet mass change. As a reference, the entire East Antarctica, about four times larger in area than the west, showed a total relative mass change of less than 500 Gt according to the datasets from (6) (gray curve in Fig. 1F) and (7) (gray curve in Fig. 1G).

There is strong consistency among reconstructed snow accumulation, ice core records, and snowfall data in the two reanalysis datasets. The consistency highlights the robustness of the dipole-like pattern of snowfall trends, which causes a total increase in snow accumulation of more than 1500 Gt over West Antarctica in the entire 20th century. According to the reanalysis data, these trends are accompanied by a strengthened ASL against the backdrop of an intensified SAM throughout the 20th century (fig. S5). Recent studies (24, 28, 38, 39) indicate that atmospheric circulation adjustments may mediate the effects of the tropical oceans on polar climate variability on interannual or decadal timescales. Despite the potential bias in the long-term trends in the reanalysis datasets, the process on interannual and decadal timescales motivates us to investigate whether similar atmospheric teleconnection mechanisms that operate on centennial timescales contribute to the contrasting snow accumulation changes over West Antarctica.

Contrasting 20th century tropical SST trends between the Pacific and the Atlantic

Since 1901, most areas of the tropical ocean have undergone continuous warming (Fig. 2A), with the most pronounced changes observed in the tropical Atlantic, contrasted by a statistically insignificant cooling trend over the equatorial central-eastern Pacific (32, 34, 40). Considering the critical role of relative SST (41), defined as the deviation of the local SST from the tropical mean temperature, in driving deep atmospheric convection and subsequent teleconnection patterns, we evaluate the changes in the relative SST across the tropics (Fig. 2B). Results show a clearer relative cooling in the central and eastern tropical Pacific and a relative warming in the tropical Atlantic.

We calculate the spatial mean SST time series (Fig. 2C) for both the equatorial Pacific (blue box in Fig. 2A) and the tropical Atlantic (red box in Fig. 2A), along with the area-weighted mean SST index for the entire tropics. Throughout the 20th century, SST changes in the tropical Atlantic were almost always ~ 0.2 K higher than the tropical mean, while the equatorial Pacific was ~ 0.2 K lower than the tropical mean (Fig. 2C). Given that the long-term changes of the mean temperature over the tropical oceans may also be influenced by the greenhouse warming trend on centennial timescales, the changes in the tropical convective precipitation and thus the teleconnection patterns are more associated with the relative SST over tropics rather than the SST. We calculate the changes in the relative SST across the tropics (Fig. 2, C and D), results showing that the tropical Pacific got

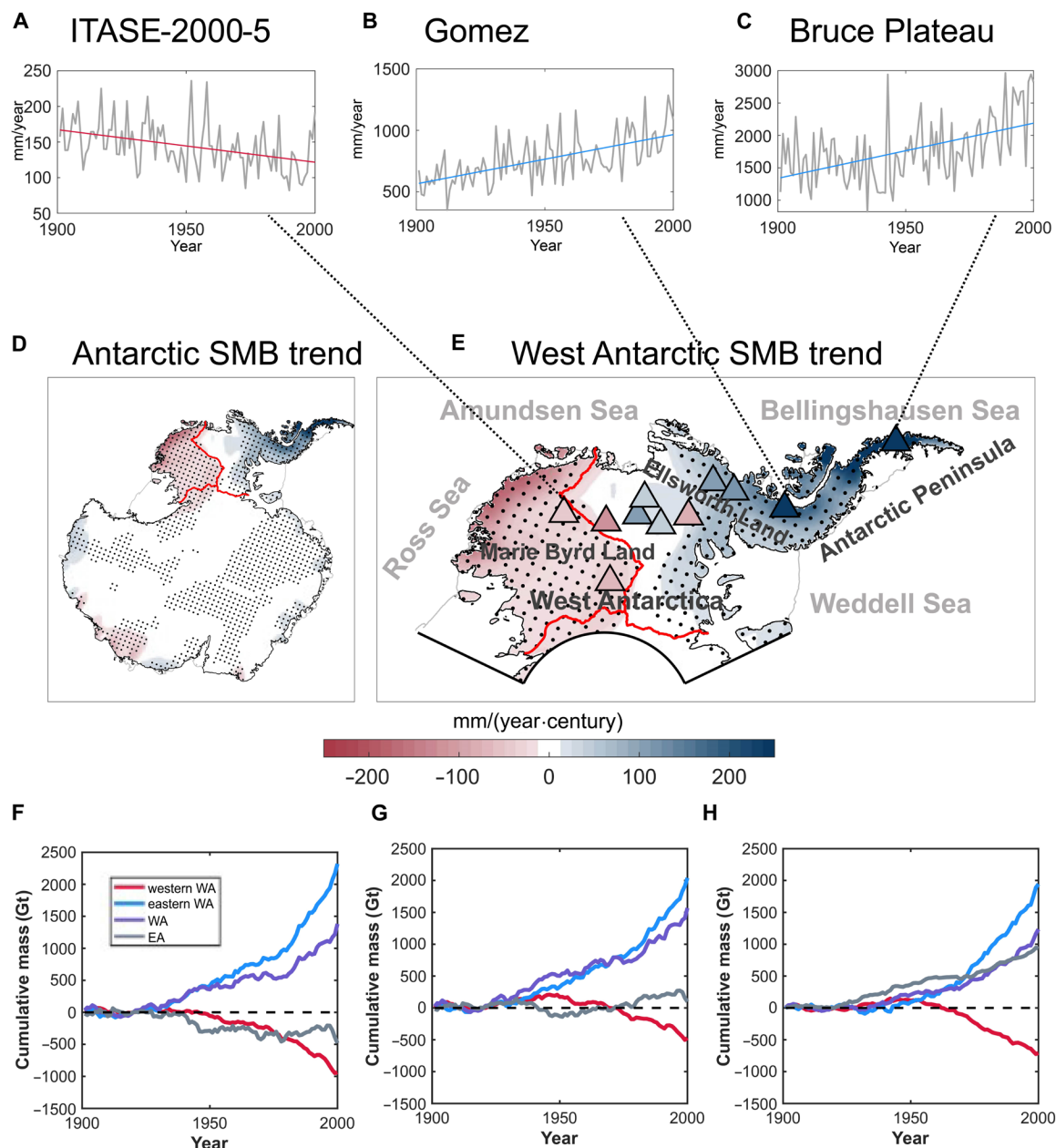


Fig. 1. Centennial trends in snow accumulation rate during the 20th century. (A to C) show the time series of snow accumulation rate of three ice cores: ITASE-2000-5 (A), Gomez (B), and Bruce Plateau (C). The positive (negative) trend in each ice core is illustrated by the blue (red) line in (A) to (C). Panel (D) shows the centennial trend of Antarctic snow accumulation rate using a reconstruction dataset (6), and panel (E) shows the trend in West Antarctica. Black dots denote the area where the trends are statistically significant at the 90% level. Triangles in (E) show all ice core records with significant negative (red) and positive (blue) snow accumulation trends. The red lines in (D) and (E) denote the boundary of three Antarctic sectors: western West Antarctica, eastern West Antarctica, and East Antarctica. Panels (F) to (H) show the cumulative mass changes during the entire 20th century relative to the mean value of snow accumulation rate during the 1881–1920 period (see “Cumulative mass change” in Materials and Methods) using reconstruction dataset from (6) (F), (7) (G), and (8) (H), respectively. In the 20th century, the snow accumulation rate over West Antarctica markedly increased, whose spatial pattern shows an eastern-western contrast feature with the Ellsworth Land and the Peninsula dominates the increase trend.

relatively cooler in the 20th century, whereas the tropical Atlantic got relatively warmer. The time series of the relative SST over the equatorial Pacific and the tropical Atlantic reemphasizes these contrasting SST changes between the two basins, potentially influencing climate changes over the Southern Hemisphere through atmosphere-ocean interactions and Rossby wave dynamics (24, 38, 39).

Linkage between tropical SST and West Antarctic snow accumulation

To explore the relationship between the Antarctic snow accumulation rate and both detrended tropical SST and detrended tropical relative SST, we conduct maximized covariance analysis (MCA) (fig. S6) (see the “MCA analysis” in Materials and Methods). The first

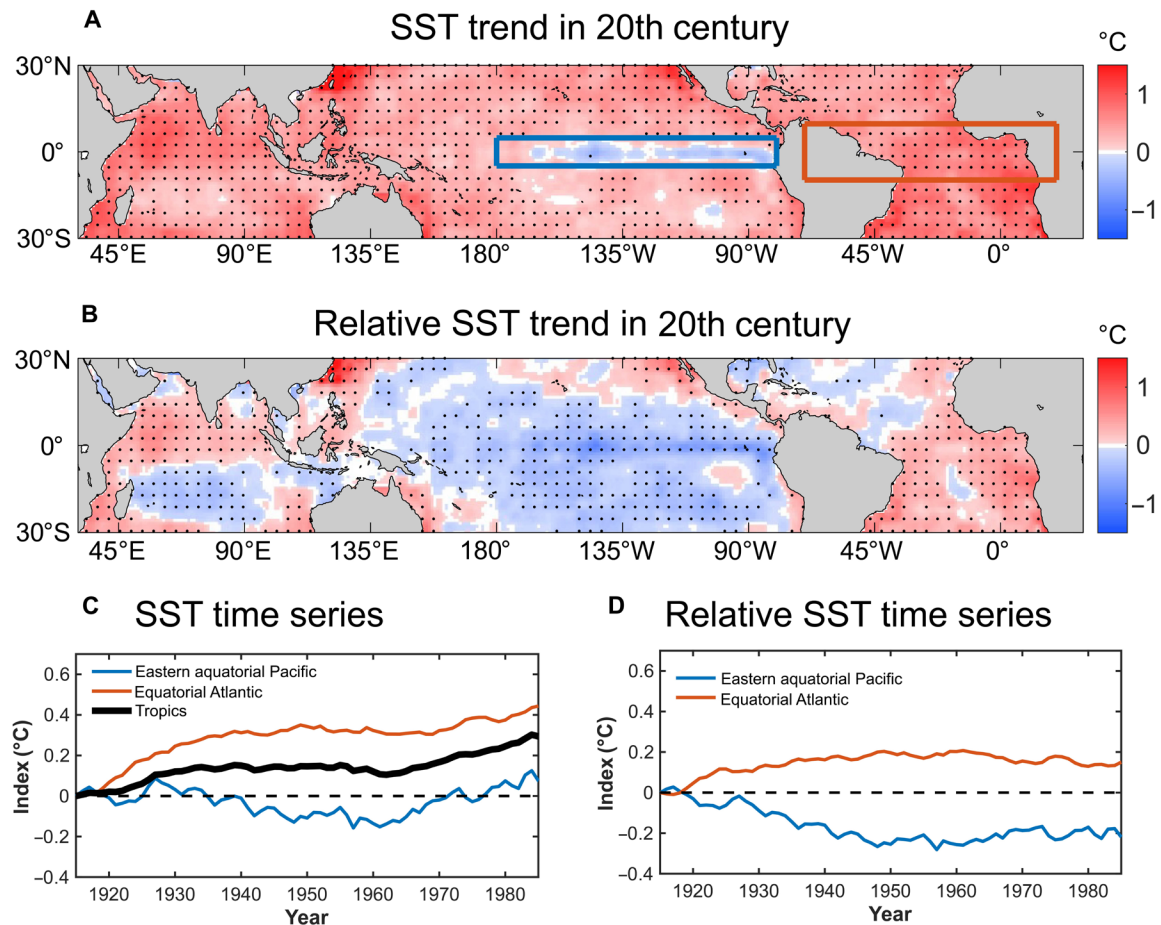


Fig. 2. Changes in tropical SST and relative SST during the 20th century. (A and B) Linear trends in tropical SST (A) and relative SST (B), which is defined as the deviation of the SST over each grid point from the area-weighted mean SST over the tropical ocean (30°N to 30°S) from 1901 to 2000. Black dots denote the area where the trends are statistically significant at the 90% level with the Mann-Kendall test. The blue box (180°W to 80°W, 5°N to 5°S) in A indicates the domain over which the SST is averaged to obtain the equatorial eastern Pacific SST index and the red box (10°N to 10°S) for the tropical Atlantic SST index. The SST climatology is defined on the basis of the mean value of 1891–1910. (C) Evolution of the tropical Atlantic SST [red line; defined as the red box in (A)], the equatorial Pacific SST [blue line; defined as the blue box in (A)], and the mean SST over the entire tropics (black line). A 30-year moving average is applied to the SST time series. (D), as in (C), but for the relative SST over the tropical Atlantic (red) and Pacific (blue). In the 20th century, the tropical Atlantic experiences a pronounced warming relative to the tropical mean, while the equatorial central-eastern Pacific experiences a relative cooling.

mode of each decomposition exhibits a dipole-like pattern in snow accumulation rate over West Antarctica (fig. S6, A and D), accompanied by a cooling pattern over the tropical central-eastern Pacific and warming signals over the tropical western Pacific and the tropical Atlantic (fig. S6, C and F).

The strongest SST signals over the tropical Pacific, rather than the Atlantic, due to the dominance of ENSO in variability on inter-annual timescales. The high variability in tropical Pacific SST compared with that of the global tropical mean SST leads to a similarity pattern between the results with tropical SST (fig. S6C) and relative SST (fig. S6F). Nevertheless, the warming signal over the tropical Atlantic (fig. S6F) implies the potential important influence of these warming signals on the West Antarctic snow accumulation rate. These results show the linkage between the tropical SST and the West Antarctic snow accumulation rate. It motivates us to further evaluate the relationship between the West Antarctic accumulation rate and SST variability over the tropical Pacific and Atlantic, respectively.

We regress the West Antarctic snow accumulation rate against the detrended SST time series of both the equatorial central-eastern Pacific (180°W to 80°W, 5°N to 5°S, the area with negative SST trends, blue box in Fig. 2A) and the tropical Atlantic (10°N to 10°S, red box in Fig. 2A), respectively. Equatorial Pacific warming typically aligns with a dipole-like accumulation pattern in West Antarctica (fig. S7A), characterized by increased rates (7.8 ± 0.5 mm/year per °C) over Marie-Byrd Land and decreased rates (-7.1 ± 0.6 mm/year per °C) over Ellsworth Land and the Peninsula. Conversely, tropical Atlantic warming mainly correlates with a strong increase (20.8 ± 1.5 mm/year per °C) in accumulation rate over Ellsworth Land and the Peninsula (fig. S7D). Similar relationships are seen in the precipitation patterns over West Antarctica (fig. S8) in the re-analysis dataset.

To understand the underlying mediating effect of atmospheric circulation adjustment between tropical SST variability and the West Antarctic snow accumulation changes, we examine the circulation anomalies associated with the detrended equatorial Pacific and tropical Atlantic

SST variability (fig. S7, C and F), respectively. Equatorial Pacific warming generates an anomalous atmospheric circulation pattern (fig. S7C), similar to the Pacific South America pattern (42, 43), manifesting as an anticyclonic anomaly over the ASL area. On the other hand, tropical Atlantic warming triggers a Rossby wave train, traversing the globe across the Southern Hemisphere and deepening the ASL (cyclonic anomaly) (fig. S7F) (28, 29). The anticyclonic (cyclonic) circulation anomaly over West Antarctica associated with the equatorial Pacific (tropical Atlantic) warming can strengthen (weaken) the moisture advection toward Marie-Byrd Land and weaken (strengthen) the advection toward Ellsworth Land and the Peninsula (fig. S7, B and E). This leads to a moisture convergence (divergence) over Marie-Byrd Land but a divergence (convergence) over Ellsworth Land and the Peninsula, thus driving the dipole-like pattern of the West Antarctic snow accumulation changes (fig. S6, A and D).

SST warming over the equatorial Pacific and the tropical Atlantic drives opposite circulation and precipitation anomalies over West Antarctica. However, given that the equatorial Pacific, which shows a cooling, and the tropical Atlantic, which shows a warming, experienced opposite SST trends during the 20th century (Fig. 2A), the effects of these two ocean basins on the centennial-scale Antarctic climate changes are coherent, both deepening the ASL and driving dipole-like accumulation rate trends over West Antarctica. A similar regression analysis is conducted with tropical Indian Ocean SST variability (fig. S9). The anomalies of accumulation rate (fig. S9A) related to the tropical Indian Ocean warming are characterized by an increased snowfall over most areas, associated with enhanced moisture transport from ocean to land (fig. S9B), which is a pattern distinctly different from the above dipole-like pattern. We therefore focus on the effects of the tropical Pacific and the Atlantic Oceans.

Simulation results of century-long teleconnection

Our statistical analysis reveals the linkages between tropical SST variability and the Antarctic snow accumulation changes, but a causal relationship is yet to be established. To determine this causality, we conduct atmospheric model experiments, forced by the SST trends over the tropical Pacific and Atlantic from 1901 to 2000, respectively (see “Atmospheric model and experiments design” in Materials and Methods). Simulation results strongly reinforce the idea that both the centennial cooling over the equatorial Pacific and the warming over the tropical Atlantic deepen the ASL (Fig. 3, C and F), driving a cyclonic circulation anomaly and subsequent moisture advection around West Antarctica (Fig. 3, B and E), resulting in dipole-like changes in the West Antarctic snow accumulation rate (Fig. 3, A and D).

Modeling results indicate a deepening in sea level pressure of ~0.5 hPa over ASL regions (contours in Fig. 3C) (44) forced by tropical Pacific SST trends during the 20th century. Concurrently, tropical Atlantic SST trends lead to a deepening in sea level pressure of ~1.1 hPa over ASL regions (contours in Fig. 3F) during the same period. To investigate the combined influence of tropical Pacific and Atlantic oceans on the West Antarctic, we conduct model experiments forced by the SST trends over the tropical Pacific and Atlantic oceans simultaneously. The patterns of the changes in atmospheric circulation and precipitation are largely similar to the summed responses to tropical Pacific SST forcing and tropical Atlantic SST forcing, but with a slightly weaker magnitude (~10%), manifesting a deepening in sea level pressure of ~1.5 hPa over ASL regions (contours in fig. S10A).

Considering that the stationary Rossby wave trains linking the tropics to the polar regions are usually triggered by the tropical deep atmospheric convection, which is associated with the relative SST (41), we conduct additional model experiments forced by the changes in the relative SST over the tropical Pacific and the Atlantic Oceans, respectively (fig. S11) (see the “Atmospheric model and experiments design” in Materials and Methods). The results show that the relative SST changes over both the Pacific and the Atlantic drive stationary Rossby wave trains to the high-latitude Southern Hemisphere, thereby deepening the ASL and leading to the dipole-like trends in the West Antarctic snow accumulation.

DISCUSSION

By combining centennial-scale observations, reanalysis data, and numerical model experiments, we reveal a teleconnection between centennial tropical SST trends and Antarctic snow accumulation and surface mass balance. The converging result from statistical analysis, observed trend patterns, simulations, and atmospheric dynamical theory, highlights the robustness of this teleconnection. Through the teleconnection, contrasting SST trends over the tropical Pacific and Atlantic Oceans since the early 20th century coherently drove an adjustment in moisture transport around the Amundsen and neighboring seas through Rossby wave dynamics (Fig. 4), leading to the dipole-like snowfall trend pattern over West Antarctica. The pattern substantially contributed to the surface mass balance of the Antarctic Ice Sheet during the 20th century.

Thus, our study extends previous research to centennial timescales by combining the historical tropical SST measurements with the Antarctic ice core records. This is an unexpected result, as it is not all obvious a priori that the relatively well-understood interannual and decadal behavior of tropical-polar atmospheric teleconnections operate on the centennial timescale. The longer, centennial timescale studied here brings into consideration some complicating factors not present on the shorter interannual or decadal time scales. For one, the relative importance of the tropical Pacific versus Atlantic on the increased snow accumulation in Antarctica had been unclear before our current analysis. Moreover, the centennial-scale trend of different tropical ocean basins may also interact with each other (34), thus forming an interactive system between the tropics and the high latitudes.

It is important to notice that the teleconnection pattern revealed in this study demonstrate the remote effect of the Pacific cooling and Atlantic warming qualitatively. However, it is difficult to accurately quantify the remote effect on the centennial changes in the Antarctic surface mass balance based on these regression and simulation results. On interannual timescales, the atmospheric teleconnection is generated by a perturbation relative to the background state, while the centennial changes represent a drift of the background state. This climate drift of the tropical SST may induce a large-scale adjustment of the tropical convective motion and further an adjustment of the global atmospheric background state, which may influence the phases and amplitude of the atmospheric teleconnection pattern. Moreover, on centennial timescales, we have to take the greenhouse gas warming into account, which makes physical process even more complicated. The relative SST helps to evaluate the effect of the tropical SST gradient under global warming. However, given such a complicated nonlinear system, it is still difficult to accurately quantify the remote effect of this tropical climate drift. This will be a main focus of our future study.

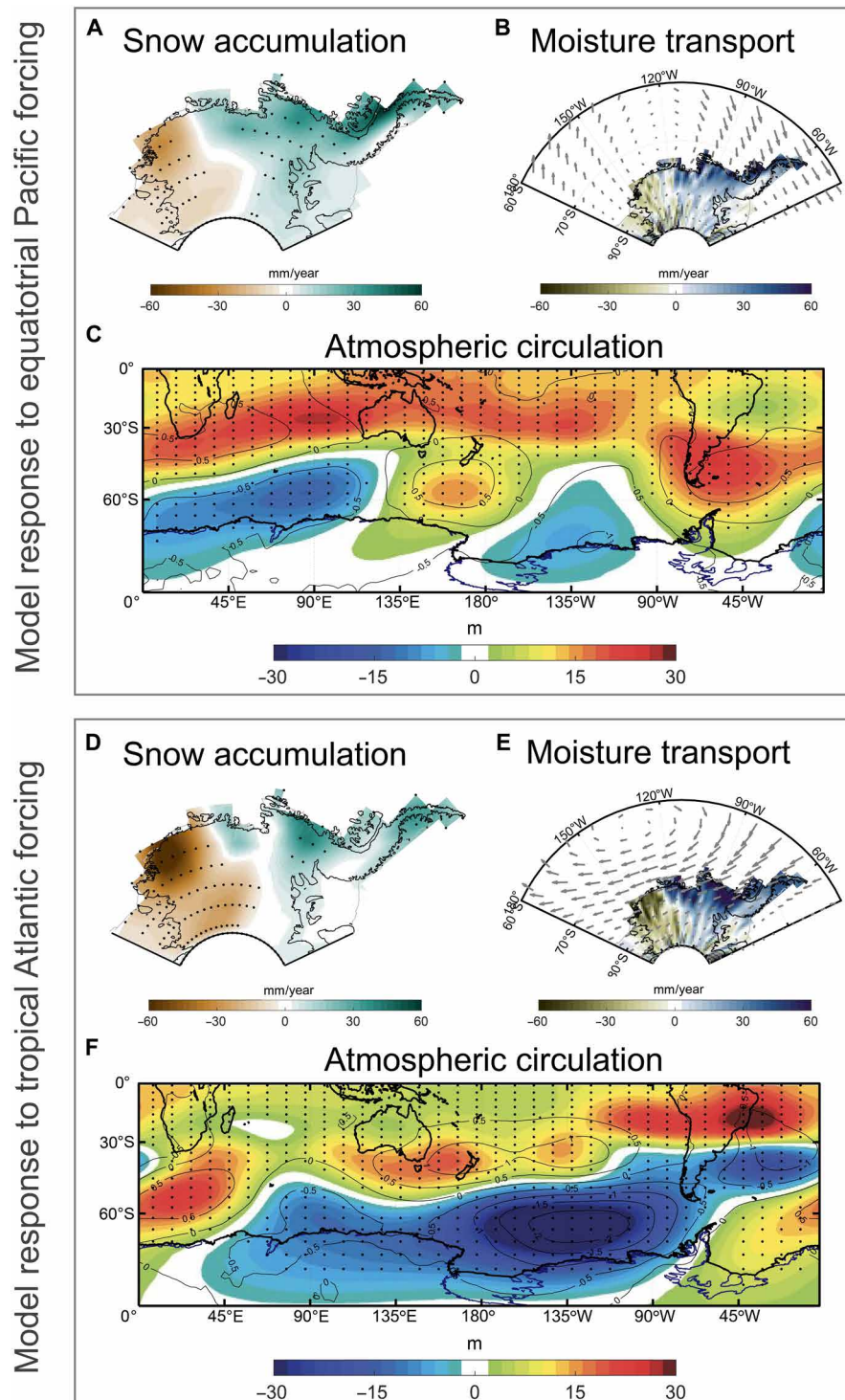


Fig. 3. Simulated response to the observed trends in tropical Pacific and Atlantic SST during the 20th century. (A to C) Model response of precipitation (A), moisture transport [vectors in (B)] and its divergence (color shading in (B)), sea level pressure [contour in (C), with an interval of 0.5 hPa], and 200-hPa geopotential height [color shading in (C)] to the tropical Pacific SST forcing (see Materials and Methods). (D to F) Same as (A) to (C), but for the response to the tropical Atlantic SST forcing. Black dots denote the areas with statistically significant level of >90% based on the two-sided and paired Student's *t* test. Simulation results reinforce the idea that both the centennial cooling over the equatorial Pacific and the warming over the tropical Atlantic deepen the ASL, resulting in dipole-like changes in the West Antarctic snow accumulation rate.

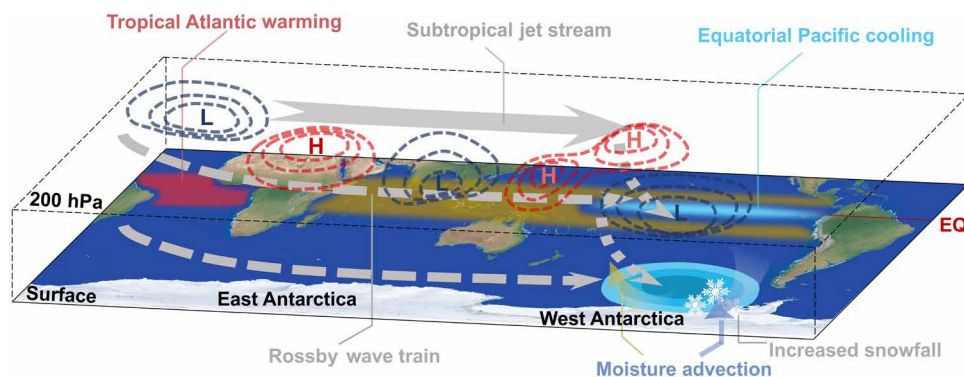


Fig. 4. Schematic of centennial teleconnection linking the tropical SST trends and the snow accumulation changes over West Antarctica. During the 20th century, the tropical Atlantic has experienced the most pronounced warming, contrasting with an insignificant cooling trend in the equatorial central-eastern Pacific. Both the Pacific and the Atlantic SST changes drive a cyclonic anomaly over the ASL region through Rossby wave dynamics. These two teleconnections collectively induce moisture advection, leading to increased snowfall in eastern West Antarctica and decreased snowfall in its western part.

Our results emphasize the need for better understanding interactions on centennial-scale climate change between different regions of the Earth system. Continued anthropogenic emission of carbon dioxide and methane are likely to produce relatively uniform climate forcing (45, 46), while ozone and aerosols may result in spatially varied forcing (46, 47). On the other hand, centennial teleconnections may create strong differences in climate change between different regions. Atmospheric dynamics (24, 29, 39) and atmosphere-ocean interactions (48) ultimately dictate how climate is changed under greenhouse gas forcing, involving modulating teleconnections that drive them to produce climate change trends, and highlighting an urgent need for further research to decode these processes, the timescales on which they operate, and assess their regional and global impacts.

MATERIALS AND METHODS

Datasets

We use the Met Office Hadley Centre sea ice and SST (HadISST) dataset (37) to evaluate the trends and evolution of the tropical SST and to force the atmospheric model below. The datasets extend back to 1870. However, because of the reported reliability of observations post-1900 (37), our analysis is confined to the 20th century SST data. Before satellite observations (post-1980s), HadISST relied on ship-based and other in situ measurements, which were sparser in earlier periods, leading to greater uncertainties. In addition, middle and low latitudes have more observations compared to polar regions (37, 49).

We used three reconstructed snow accumulation datasets from (6), (7), and (8). The dataset from (6) covers both the 19th and 20th centuries. The data are derived by synthesizing a compile of ice-core measured snow accumulation using spatial patterns from reanalysis datasets, European Centre for Medium-Range Weather Forecasts “Interim” (ERA-interim), Modern-Era Retrospective Analysis for Research and Applications Version 2 (MERRA2), and Climate Forecast System Reanalysis (CFSR). The results based on three reanalysis datasets are similar (not shown), we only show the analysis result using reconstruction data based on ERA-interim. The 1-sigma uncertainty in the reconstructed snow accumulation from (6) over the coastal region of West Antarctica is 50 to 100 mm/year, ~200 mm/year over the high precipitation area in Antarctica Peninsula, and ~25 mm/year for the interior area of West and East Antarctica. The dataset from (7)

also combines ice core records with outputs of climate models. They reconcile five reanalysis products and two regional climate models, and the assembled mean of the seven independent products provides a reconciled 310-year reconstruction of Antarctic surface mass balance. The dataset from (8) is reconstructed through data assimilation. An article filter method is used to assimilate observations, ice core snow accumulation records, tree ring width records, and auto weather station data into a coupled model, isotope-enabled iCESM1. The 1-sigma uncertainty in the reconstructed snow accumulation from (6) over the coastal region of West Antarctica and Peninsula are <100 mm/year and ~20 mm/year for the interior area of West and East Antarctica. To align with the periods of the SST observations, our analysis focuses exclusively on the 20th century snow accumulation data to evaluate the relationship between the centennial-scale change in tropical SST and Antarctic surface mass balance.

In addition, surface mass balance records from 20 ice cores are also used to verify the centennial trends of snow accumulation rate across West Antarctica. More details about these ice cores are provided in table S1. The primary source of uncertainty in these records arises from the dating of the ice cores, particularly in the interior of Antarctica where snow accumulation rates are so low that glaciological noise can overwhelm the interannual variability (50). However, this uncertainty is somewhat mitigated when examining the multidecadal and the centennial trends.

The net precipitation (precipitation minus evaporation) derived from atmospheric reanalysis well represents the snow accumulation over Antarctica (6). Precipitation and evaporation from two state-of-the-art datasets, NOAA-CIRES-DOE 20th Century Reanalysis version 3 (20CrV3) (51) and ECMWF’s atmospheric reanalysis of the 20th century (ERA-20C) (52) are used to evaluate and validate the changes in snow accumulation rate in West Antarctica. Other variables of these two datasets, including the 200- and 500-hPa geopotential height, sea level pressure, wind, and specific humidity from both datasets are also used to examine the changes in atmospheric circulation and the corresponding adjustments in moisture transport around West Antarctica.

Cumulative mass change

We follow approaches in (6) to evaluate the cumulative mass change attributable to snow accumulation rate changes. Basin division is

defined in (53). Basins 1 and 21 to 26 are combined to represent changes in eastern West Antarctica, and basins 18 to 20 represent changes in western West Antarctica. Basins 2 to 17 are combined to evaluate changes in East Antarctica. Area weighting is considered during the integration of accumulation data.

According to the reconstruction data, West Antarctica saw a centennial increased trend in the east but decreased trend in the west. So, we choose the first 20 years (1901–1920) to represent the climatology of accumulation rate in the early 20th century and cumulate the difference between the annual accumulation rate and 1901–1920 mean value on each grid and then integrate them by area weighting to assess the additional snow accumulation due to the persistent increase or decrease during the 20th century. The additional snow accumulation is referred to as cumulative mass change because it contributes to the Antarctic Ice Sheet mass change over the past century. The results of the climatology calculations based on different time period selections are presented in fig. S4.

Trend analysis

Sen's slope method (54) is conducted to retrieve the trends in the West Antarctic snow accumulation rate from reconstructions and ice cores, precipitation, and evaporation from reanalysis data. Mann-Kendall test is used to assess the statistical significance (54). The trend analysis is conducted using the reconstruction and reanalysis dataset's original resolutions. Reconstructed snow accumulation rate trends during the 20th century are calculated to retrieve the centennial changes in the surface mass balance of West Antarctica, verified by the accumulation rate trends from the ice cores. Only the significant trend from ice cores with a confidence level of at least 90% is shown in Fig. 1. Trends in the annual precipitation and evaporation (1901–2000) from both 20CRv3 and ERA-20C (fig. S2) are calculated to corroborate the centennial changes in snow accumulation revealed by the reconstruction data.

Using Sen's slope method, centennial trends from 1901–2000 of tropical SST and relative SST are calculated at each grid point of annual mean SST from HadISST (Fig. 2). The relative SST is calculated by subtracting the tropical mean SST from each grid point and each time step. The annual mean 500 hPa geopotential height trends during the 20th century from both datasets (fig. S5) are calculated by ordinary least squares regression to show the centennial deepening of the SAM and ASL. Mann-Kendall test is also used to assess the statistical significance. The two reanalysis data both indicate a strengthened ASL (-5.1 ± 1.8 hPa in 20CRv3 and -9.4 ± 3.5 hPa in ERA-20C) (44, 55) against the backdrop of an intensified SAM throughout the 20th century. Considering the underlying bias of long-term trends in the atmospheric circulation from the reanalysis datasets (31, 51), the centennial-scale deepening of the ASL may be overestimated. Therefore, it primarily serves as a clue to investigate the mechanism behind the centennial teleconnection between the tropical SST and the West Antarctic snow accumulation.

Maximized covariance analysis

To identify the covariability between snow accumulation in West Antarctica and tropical SST, we used the MCA. This analysis is applied to two sets of data: first, the annual mean accumulation rate in West Antarctica from reconstruction data and SST in the tropics from HadISST, and second, the same West Antarctic snow accumulation rate and the relative SST in the tropics. Before conducting MCA, linear trends are removed from each grid in the tropical SST and the accumulation rate data in West Antarctica.

The MCA results describe leading modes that have the maximum covariance and dominate variability in both fields through a singular value decomposition. The first mode of MCA explains a significant portion of the covariance, accounting for 81% with tropical SST and 84% with relative SST. In the MCA results, the correlation coefficients between the tropical SST and West Antarctic snow accumulation time series are 0.37 and 0.40, respectively, for the tropical SST and relative SST modes, both significant at the <1% level. The MCA results reveal the strong covariability between the tropical SST and the dipole-like accumulation rate pattern in West Antarctica on interannual, decadal, and multidecadal timescales, which motivate us to investigate their relationship on centennial timescale.

Regression analysis

Ordinary least squares regression is used to evaluate the individual contribution of SST in the tropical Pacific, Atlantic and Indian Oceans to the snow accumulation and associated circulation anomaly over West Antarctica. The equatorial Pacific and tropical Atlantic SST index are defined as the area-weighted annual mean SST in the blue and red boxes, respectively, in fig. S7A. The tropical Indian Ocean index is defined as the area-weighted mean SST within the region 20°S to 20°N and 40°E to 110°E. All variables are detrended before analysis.

The accumulation rate from reconstruction data and precipitation from 20CRv3 are regressed against the equatorial Pacific and the tropical Atlantic SST index to evaluate the influence on surface mass balance from each basin (figs. S7, A and D; S8; and S9A). A total of 200 hPa geopotential height and sea level pressure from 20CRv3 are regressed to evaluate the associated circulation changes (figs. S6, C and F and S8C). Moisture transports are calculated by vertically integrating the horizontal moisture transport at each level based on the following formula

$$\text{moisture transport} = \frac{1}{g} \int_{p_t}^{p_s} (qV)' dp \quad (1)$$

where V is the horizontal wind velocity at each level, q is the specific humidity, p_s is the surface pressure, p_t is the top pressure (10 hPa), and g is the gravitation acceleration. Moisture transports and associated divergence are regressed to evaluate the moisture advection aroused by the circulation adjustment (figs. S6, B and E and S8B). All variables are detrended before regression analysis.

Confidence intervals for the regression coefficients are determined via two-sided and unpaired Student's t test, and autocorrelation is accounted for by adjusting the degrees of freedom as follows (56)

$$T^* = T(1 - r_1 r_2) / (1 + r_1 r_2) \quad (2)$$

where T^* and T are the adjusted and unadjusted degrees of freedom, r_1 and r_2 are the lag-one autocorrelation of the respective time series.

Atmospheric model and experiments design

Community Atmosphere Model version 5, a global atmosphere model developed by the National Center for Atmospheric Research, is used to validate and evaluate the contribution of the centennial trend in Atlantic and Pacific SST to snow accumulation changes in West Antarctica. We use a global horizontal resolution of $1.9^\circ \times 2.5^\circ$ finite volume grid. We use the Community Land Model and the

thermodynamic module of the Community Sea-Ice Model to generate the heat flux at the surface. The atmospheric model is solely forced by the SST as its external lower boundary forcing.

A control run and four forced runs are conducted to assess the response to both raw and relative-to-tropical-mean trends in the tropical Pacific and Atlantic over the 20th century. For the control run, the model is forced by the climatology SST and sea ice from 1891 to 1910 with an annual cycle, while all other forcings—including solar radiation, greenhouse gas concentrations, and aerosols—are fixed at the mean levels in the 20th century. In the two forced runs for the Pacific and Atlantic raw SST, 20th century tropical SST (20°S to 20°N) trends for each basin are superimposed onto the climatology SST. In the two relative SST runs, the tropical mean SST trends are subtracted from the SST forcings.

Each experiment is integrated for 130 years, and the mean values of the past 100 years are calculated for subsequent analysis. Differences between the results of the forced runs and the control run are calculated to assess the response to the centennial trends in SST from each tropical ocean basin. Two-sided and paired Student's *t* test are used to determine the significance. The integration duration of 100 years is chosen to adequately extract the response from natural atmospheric variability.

We conduct additional experiments using the mean value of the global SST during 1901–1920 as the climatology SST (fig. S12). Results show similar changes in atmospheric circulation, moisture transport, and precipitation as in the previous experiments (Fig. 3).

Supplementary Materials

This PDF file includes:

Figs. S1 to S12

Table S1

References

REFERENCES AND NOTES

1. E. Rignot, J. Mouginot, B. Scheuchl, M. R. van den Broeke, M. van Wessem, M. Morlighem, Four decades of Antarctic Ice Sheet mass balance from 1979–2017. *Proc. Natl. Acad. Sci. U.S.A.* **116**, 1095–1103 (2019).
2. A. Shepherd, E. Ivins, E. Rignot, B. Smith, M. van den Broeke, I. Velicogna, P. Whitehouse, K. Briggs, I. Joughin, G. Krinner, S. Nowicki, T. Payne, T. Scambos, N. Schlegel, A. Geruo, C. Agosta, A. Ahlström, G. Babonis, V. Barletta, A. Blazquez, J. Bonin, B. Csatho, R. Cullather, D. Felikson, X. Fettweis, R. Forsberg, H. Gallee, A. Gardner, L. Gilbert, A. Groh, B. Gunter, E. Hanna, C. Harig, V. Helm, A. Horvath, M. Horwath, S. Khan, K. K. Kjeldsen, H. Konrad, P. Langen, B. Lecavalier, B. Loomis, S. Luthcke, M. M. Millan, D. Melini, S. Mernild, Y. Mohajerani, P. Moore, J. Mouginot, G. Moyano, A. Muir, T. Nagler, G. Nield, J. Nilsson, B. Noel, I. Otosaka, M. E. Pattle, W. R. Peltier, N. Pie, R. Rietbroek, H. Rott, L. Sandberg-Sørensen, I. Sasgen, H. Save, B. Scheuchl, E. Schrama, L. Schröder, K.-W. Seo, S. Simonsen, T. Slater, G. Spada, T. Sutterley, M. Talpe, L. Tarasov, W. J. van de Berg, W. van der Wal, M. van Wessem, B. D. Vishwakarma, D. Wiese, B. Wouters, The IMBIE team, Mass balance of the Antarctic Ice Sheet from 1992 to 2017. *Nature* **558**, 219–222 (2018).
3. S. K. Gulev, P. W. Thorne, J. Ahn, F. J. Dentener, C. M. Domingues, S. Gerland, D. Gong, D. S. Kaufman, H. C. Nnamchi, J. Quaas, J. A. Rivera, S. Sathyendranath, S. L. Smith, B. Trewin, K. von Schuckmann, R. S. Vose, “Changing State of the Climate System” in *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*, V. Masson-Delmotte, P. Zhai, A. Pirani, S. L. Connors, C. Péan, S. Berger, N. Caud, Y. Chen, L. Goldfarb, M. I. Gomis, M. Huang, K. Leitzell, E. Lonnoy, J. B. R. Matthews, T. K. Maycock, T. Waterfield, O. Yelekçi, R. Yu, B. Zhou, Eds. (Cambridge University Press, 2021), pp. 287–422.
4. M. A. Depoorter, J. L. Bamber, J. A. Griggs, J. T. M. Lenaerts, S. R. M. Ligtienberg, M. R. van den Broeke, G. Moholdt, Calving fluxes and basal melt rates of Antarctic ice shelves. *Nature* **502**, 89–92 (2013).
5. Y. Liu, J. C. Moore, X. Cheng, R. M. Gladstone, J. N. Bassis, H. Liu, J. Wen, F. Hui, Ocean-driven thinning enhances iceberg calving and retreat of Antarctic ice shelves. *Proc. Natl. Acad. Sci. U.S.A.* **112**, 3263–3268 (2015).
6. B. Medley, E. R. Thomas, Increased snowfall over the Antarctic Ice Sheet mitigated twentieth-century sea-level rise. *Nat. Clim. Change* **9**, 34–39 (2019).
7. Y. Wang, C. Xiao, An increase in the Antarctic surface mass balance during the past three centuries, dampening global sea level rise. *J. Climate* **36**, 8127–8138 (2023).
8. Q. Dalaiden, H. Goosse, J. Rezsőhazy, E. R. Thomas, Reconstructing atmospheric circulation and sea-ice extent in the West Antarctic over the past 200 years using data assimilation. *Climate Dynam.* **57**, 3479–3503 (2021).
9. Q. Dalaiden, A. P. Schurer, M. C. Kirchmeier-Young, H. Goosse, G. C. Hegerl, West Antarctic surface climate changes since the Mid-20th century driven by anthropogenic forcing. *Geophys. Res. Lett.* **49**, e2022GL099543 (2022).
10. E. R. Thomas, J. M. van Wessem, J. Roberts, E. Isaksson, E. Schlosser, T. J. Fudge, P. Vallelonga, B. Medley, J. Lenaerts, N. Bertler, M. R. van den Broeke, D. A. Dixon, M. Frezzotti, B. Stenni, M. Curran, A. A. Ekaykin, Regional Antarctic snow accumulation over the past 1000 years. *Clim. Past* **13**, 1491–1513 (2017).
11. A. Shepherd, H. A. Fricker, S. L. Farrell, Trends and connections across the Antarctic cryosphere. *Nature* **558**, 223–232 (2018).
12. D. H. Bromwich, J. P. Nicolas, A. J. Monaghan, M. A. Lazzara, L. M. Keller, G. A. Weidner, A. B. Wilson, Central West Antarctica among the most rapidly warming regions on Earth. *Nat. Geosci.* **6**, 139–145 (2013).
13. N. Yuan, M. Ding, Y. Huang, Z. Fu, E. Xoplaki, J. Luterbacher, On the long-term climate memory in the surface air temperature records over Antarctica: A nonnegligible factor for trend evaluation. *J. Climate* **28**, 5922–5934 (2015).
14. D. M. Holland, K. W. Nicholls, A. Basinski, The Southern Ocean and its interaction with the Antarctic Ice Sheet. *Science* **367**, 1326–1330 (2020).
15. I. Joughin, R. B. Alley, Stability of the West Antarctic ice sheet in a warming world. *Nat. Geosci.* **4**, 506–513 (2011).
16. K. A. Naughten, P. R. Holland, J. De Rydt, Unavoidable future increase in West Antarctic ice-shelf melting over the twenty-first century. *Nat. Clim. Change* **13**, 1222–1228 (2023).
17. K. I. Ohshima, Y. Fukamachi, G. D. Williams, S. Nihashi, F. Roquet, Y. Kitade, T. Tamura, D. Hirano, L. Herráiz-Borreguero, I. Field, M. Hindell, S. Aoki, M. Wakatsuchi, Antarctic Bottom Water production by intense sea-ice formation in the Cape Darnley polynya. *Nat. Geosci.* **6**, 235–240 (2013).
18. N. R. Golledge, E. D. Keller, N. Gomez, K. A. Naughten, J. Bernalles, L. D. Trusel, T. L. Edwards, Global environmental consequences of twenty-first-century ice-sheet melt. *Nature* **566**, 65–72 (2019).
19. S. Sadai, A. Condron, R. DeConto, D. Pollard, Future climate response to Antarctic Ice Sheet melt caused by anthropogenic warming. *Sci. Adv.* **6**, eaaz1169 (2020).
20. J. W. B. Rae, A. Burke, L. F. Robinson, J. F. Adkins, T. Chen, C. Cole, R. Greenop, T. Li, E. F. M. Little, D. C. Nita, J. A. Stewart, B. J. Taylor, CO₂ storage and release in the deep Southern Ocean on millennial to centennial timescales. *Nature* **562**, 569–573 (2018).
21. X. Li, W. Cai, G. A. Meehl, D. Chen, X. Yuan, M. Raphael, D. M. Holland, Q. Ding, R. L. Fogt, B. R. Markle, G. Wang, D. H. Bromwich, J. Turner, S.-P. Xie, E. J. Steig, S. T. Gille, C. Xiao, B. Wu, M. A. Lazzara, X. Chen, S. Stammerjohn, P. R. Holland, M. M. Holland, X. Cheng, S. F. Price, Z. Wang, C. M. Bitz, J. Shi, E. P. Gerber, X. Liang, H. Goosse, C. Yoo, M. Ding, L. Geng, M. Xin, C. Li, T. Dou, C. Liu, W. Sun, X. Wang, C. Song, Tropical teleconnection impacts on Antarctic climate changes. *Nat. Rev. Earth Environ.* **2**, 680–698 (2021).
22. Q. Ding, E. J. Steig, D. S. Battisti, J. M. Wallace, Influence of the Tropics on the Southern Annular Mode. *J. Climate* **25**, 6330–6348 (2012).
23. G. A. Meehl, J. M. Arblaster, C. M. Bitz, C. T. Y. Chung, H. Teng, Antarctic sea-ice expansion between 2000 and 2014 driven by tropical Pacific decadal climate variability. *Nat. Geosci.* **9**, 590–595 (2016).
24. Q. Ding, E. J. Steig, Temperature change on the Antarctic Peninsula linked to the tropical Pacific. *J. Climate* **26**, 7570–7585 (2013).
25. X. Yuan, D. G. Martinson, The Antarctic dipole and its predictability. *Geophys. Res. Lett.* **28**, 3609–3612 (2001).
26. M. Nuncio, X. Yuan, The influence of the Indian ocean dipole on Antarctic sea ice. *J. Climate* **28**, 2682–2690 (2015).
27. K. Clem, R. Fogt, South Pacific circulation changes and their connection to the tropics and regional Antarctic warming in austral spring, 1979–2012. *J. Geophys. Res. Atmos.* **120**, 2773–2792 (2015).
28. X. Li, E. P. Gerber, D. M. Holland, C. Yoo, A Rossby Wave Bridge from the Tropical Atlantic to West Antarctica. *J. Climate* **28**, 2256–2273 (2015).
29. X. Li, D. M. Holland, E. P. Gerber, C. Yoo, Impacts of the north and tropical Atlantic Ocean on the Antarctic Peninsula and sea ice. *Nature* **505**, 538–542 (2014).
30. E.-S. Chung, S.-J. Kim, A. Timmermann, K.-J. Ha, S.-K. Lee, M. F. Stuecker, K. B. Rodgers, S.-S. Lee, L. Huang, Antarctic sea-ice expansion and Southern Ocean cooling linked to tropical variability. *Nat. Clim. Change* **12**, 461–468 (2022).
31. G. J. Marshall, R. L. Fogt, J. Turner, K. R. Clem, Can current reanalyses accurately portray changes in Southern Annular Mode structure prior to 1979? *Climate Dynam.* **59**, 3717–3740 (2022).
32. R. Seager, M. Cane, N. Henderson, D.-E. Lee, R. Abernathey, H. Zhang, Strengthening tropical Pacific zonal sea surface temperature gradient consistent with rising greenhouse gases. *Nat. Clim. Change* **9**, 517–522 (2019).

33. C. Deser, A. S. Phillips, M. A. Alexander, Twentieth century tropical sea surface temperature trends revisited. *Geophys. Res. Lett.* **37**, L10701 (2010).
34. F. Kucharski, I.-S. Kang, R. Farnetti, L. Feudale, Tropical Pacific response to 20th century Atlantic warming. *Geophys. Res. Lett.* **38**, L03702 (2011).
35. L. Dong, T. Zhou, The Indian Ocean Sea surface temperature warming simulated by CMIP5 models during the twentieth century: Competing forcing roles of GHGs and anthropogenic aerosols. *J. Climate* **27**, 3348–3362 (2014).
36. A. Solomon, M. Newman, Reconciling disparate twentieth-century Indo-Pacific ocean temperature trends in the instrumental record. *Nat. Clim. Change* **2**, 691–699 (2012).
37. N. A. Rayner, D. E. Parker, E. B. Horton, C. K. Folland, L. V. Alexander, D. P. Rowell, E. C. Kent, A. Kaplan, Global analyses of sea surface temperature, sea ice, and night marine air temperature since the late nineteenth century. *J. Geophys. Res. Atmos.* **108**, 4407 (2003).
38. G. R. Simpkins, Y. Peings, G. Magnusdottir, Pacific influences on tropical atlantic teleconnections to the Southern Hemisphere high latitudes. *J. Climate* **29**, 6425–6444 (2016).
39. G. R. Simpkins, S. McGregor, A. S. Taschetto, L. M. Ciasto, M. H. England, Tropical connections to climatic change in the extratropical Southern Hemisphere: The role of Atlantic SST trends. *J. Climate* **27**, 4923–4936 (2014).
40. S. Coats, K. B. Karnauskas, Are simulated and observed twentieth century tropical Pacific sea surface temperature trends significant relative to internal variability? *Geophys. Res. Lett.* **44**, 9928–9937 (2017).
41. N. C. Johnson, S.-P. Xie, Changes in the sea surface temperature threshold for tropical convection. *Nat. Geosci.* **3**, 842–845 (2010).
42. K. C. Mo, R. W. Higgins, The Pacific–south american modes and tropical convection during the Southern Hemisphere winter. *Mon. Weather Rev.* **126**, 1581–1596 (1998).
43. J. W. Kidson, Principal modes of Southern Hemisphere low-frequency variability obtained from NCEP–NCAR reanalyses. *J. Climate* **12**, 2808–2830 (1999).
44. J. Turner, T. Phillips, J. S. Hosking, G. J. Marshall, A. Orr, The Amundsen sea low. *Int. J. Climatol.* **33**, 1818–1829 (2013).
45. C. J. Smith, R. J. Kramer, G. Myhre, K. Alterskjær, W. Collins, A. Sima, O. Boucher, J.-L. Dufresne, P. Nabat, M. Michou, S. Yukimoto, J. Cole, D. Paynter, H. Shiogama, F. M. O'Connor, E. Robertson, A. Wiltshire, T. Andrews, C. Hannay, R. Miller, L. Nazarenko, A. Kirkevåg, D. Olivie, S. Fiedler, A. Lewinschal, C. Mackallah, M. Dix, R. Pincus, P. M. Forster, Effective radiative forcing and adjustments in CMIP6 models. *Atmos. Chem. Phys.* **20**, 9591–9618 (2020).
46. N. Bellouin, W. Davies, K. P. Shine, J. Quaas, J. Mülmenstädt, P. M. Forster, C. Smith, L. Lee, L. Regayre, G. Brasseur, N. Sudarchikova, I. Bouarar, O. Boucher, G. Myhre, Radiative forcing of climate change from the Copernicus reanalysis of atmospheric composition. *Earth Syst. Sci. Data* **12**, 1649–1677 (2020).
47. A. Modak, G. Bala, K. Caldeira, L. Cao, Does shortwave absorption by methane influence its effectiveness? *Climate Dynam.* **51**, 3653–3672 (2018).
48. W. Cai, L. Wu, M. Lengaigne, T. Li, S. McGregor, J.-S. Kug, J.-Y. Yu, M. F. Stuecker, A. Santoso, X. Li, Y.-G. Ham, Y. Chikamoto, B. Ng, M. J. McPhaden, Y. Du, D. Dommengat, F. Jia, J. B. Kattar, N. Keenlyside, C. Lin, J.-J. Luo, M. Martin-Rey, Y. Ruprich-Robert, G. Wang, S.-P. Xie, Y. Yang, S. M. Kang, J.-Y. Choi, B. Gan, G.-I. Kim, C.-E. Kim, S. Kim, J.-H. Kim, P. Chang, Pan-tropical climate interactions. *Science* **363**, eaav4236 (2019).
49. J. J. Kennedy, N. A. Rayner, C. P. Atkinson, R. E. Killick, An Ensemble data set of sea surface temperature change from 1850: The Met Office Hadley Centre HadSST4.0.0.0 data set. *J. Geophys. Res. Atmos.* **124**, 7719–7763 (2019).
50. J. T. M. Lenaerts, B. Medley, M. R. van den Broeke, B. Wouters, Observing and modeling ice sheet surface mass balance. *Rev. Geophys.* **57**, 376–420 (2019).
51. L. C. Silivinski, G. P. Compo, J. S. Whitaker, P. D. Sardeshmukh, B. S. Giese, C. McColl, R. Allan, X. Yin, R. Vose, H. Titchner, J. Kennedy, L. J. Spencer, L. Ashcroft, S. Brönnimann, M. Brunet, D. Camuffo, R. Cornes, T. A. Cram, R. Crouthamel, F. Domínguez-Castro, J. E. Freeman, J. Gergis, E. Hawkins, P. D. Jones, S. Jourdain, A. Kaplan, H. Kubota, F. L. Blancq, T.-C. Lee, A. Lorrey, J. Luterbacher, M. Maugeri, C. J. Mock, G. W. K. Moore, R. Przybylak, C. Pudmenzky, C. Reason, V. C. Slonosky, C. A. Smith, B. Tinz, B. Trewin, M. A. Valente, X. L. Wang, C. Wilkinson, K. Wood, P. Wyszyński, Towards a more reliable historical reanalysis: Improvements for version 3 of the Twentieth Century Reanalysis system. *Q. J. Roy. Meteorol. Soc.* **145**, 2876–2908 (2019).
52. P. Poli, H. Hersbach, D. P. Dee, P. Berrisford, A. J. Simmons, F. Vitart, P. Laloyaux, D. G. H. Tan, C. Peubey, J.-N. Thépaut, Y. Trémolet, E. V. Hólm, M. Bonavita, L. Isaksen, M. Fisher, ERA-20C: An Atmospheric Reanalysis of the Twentieth Century. *J. Climate* **29**, 4083–4097 (2016).
53. Z. H. Jay, M. B. Giovannetto, M. A. Beckley, J. L. Saba, Antarctic and Greenland Drainage Systems, GSFC Cryospheric Sciences Laboratory (2012); <https://earth.gsfc.nasa.gov/cryo/data/polar-altimetry/antarctic-and-greenland-drainage-systems>.
54. M. Gocic, S. Trajkovic, Analysis of changes in meteorological variables using Mann-Kendall and Sen's slope estimator statistical tests in Serbia. *Glob. Planet. Change* **100**, 172–182 (2013).
55. R. L. Fogt, A. J. Vovrosh, R. A. Langen, I. Simmonds, The characteristic variability and connection to the underlying synoptic activity of the Amundsen-Bellinghousen Seas Low. *J. Geophys. Res. Atmos.* **117**, D07111 (2012).
56. C. S. Bretherton, M. Widmann, V. P. Dymnikov, J. M. Wallace, I. Bladé, The effective number of spatial degrees of freedom of a time-varying field. *J. Climate* **12**, 1990–2009 (1999).
57. E. R. Thomas, P. F. Dennis, T. J. Bracegirdle, C. Franzke, Ice core evidence for significant 100-year regional warming on the Antarctic Peninsula. *Geophys. Res. Lett.* **36**, L20704 (2009).
58. A. J. Aristarain, R. J. Delmas, M. Stievenard, Ice-core study of the link between sea-salt aerosol, sea-ice cover and climate in the Antarctic Peninsula area. *Clim. Change* **67**, 63–86 (2004).
59. B. P. Goodwin, E. Mosley-Thompson, A. B. Wilson, S. E. Porter, M. R. Sierra-Hernandez, Accumulation variability in the Antarctic Peninsula: The role of large-scale atmospheric oscillations and their interactions. *J. Climate* **29**, 2579–2596 (2016).
60. E. R. Thomas, J. S. Hosking, R. R. Tuckwell, R. A. Warren, E. C. Ludlow, Twentieth century increase in snowfall in coastal West Antarctica. *Geophys. Res. Lett.* **42**, 9387–9393 (2015).
61. B. Medley, I. Joughin, S. B. Das, E. J. Steig, H. Conway, S. Gogineni, A. S. Criscitiello, J. R. McConnell, B. E. Smith, M. R. van den Broeke, J. T. M. Lenaerts, D. H. Bromwich, J. P. Nicolas, Airborne-radar and ice-core observations of annual snow accumulation over Thwaites Glacier, West Antarctica confirm the spatiotemporal variability of global and regional atmospheric models. *Geophys. Res. Lett.* **40**, 3649–3654 (2013).
62. J. R. Banta, J. R. McConnell, M. M. Frey, R. C. Bales, K. Taylor, Spatial and temporal variability in snow accumulation at the West Antarctic Ice Sheet Divide over recent centuries. *J. Geophys. Res. Atmos.* **113**, D23102 (2008).
63. M. Sigl, T. J. Fudge, M. Winstrup, J. Cole-Dai, D. Ferris, J. R. McConnell, K. C. Taylor, K. C. Welten, T. E. Woodruff, F. Adolphi, M. Bisiaux, J. T. Brook, C. Buizert, M. W. Caffee, N. W. Dunbar, R. Edwards, L. Geng, N. Iverson, B. Koffman, L. Layman, O. J. Maselli, K. McGwire, R. Muscheler, K. Nishiizumi, D. R. Pasteris, R. H. Rhodes, T. A. Sowers, The WAIS Divide deep ice core WD2014 chronology—Part 2: Annual-layer counting (0–31 ka BP). *Clim. Past* **12**, 769–786 (2016).
64. S. Kaspari, P. A. Mayewski, D. A. Dixon, V. B. Spikes, S. B. Sneed, M. J. Handley, G. S. Hamilton, Climate variability in West Antarctica derived from annual accumulation-rate records from ITASE firn/ice cores. *Ann. Glaciol.* **39**, 585–594 (2004).
65. P. A. Mayewski, D. A. Dixon, US International Trans-Antarctic Scientific Expedition (US ITASE) Glaciochemical Data (U.S. Antarctic Program Data Center, 2013); <https://doi.org/10.7265/N51V5BXR>.
66. N. A. N. Bertler, H. Conway, D. Dahl-Jensen, D. B. Emanuelsson, M. Winstrup, P. T. Vallerlonga, J. E. Lee, E. J. Brook, J. P. Severinghaus, T. J. Fudge, E. D. Keller, W. T. Baisden, R. C. A. Hindmarsh, P. D. Neff, T. Blunier, R. Edwards, P. A. Mayewski, S. Kipfstuhl, C. Buizert, S. Canessa, R. Dacic, H. A. Kjaer, A. Kurbatov, D. Zhang, E. D. Waddington, G. Baccolo, T. Beers, H. J. Brightley, L. Carter, D. Clemens-Sewall, V. G. Ciobanu, B. Delmonte, L. Eling, A. Ellis, S. Ganesh, N. R. Golledge, S. Haines, M. Handley, R. L. Hawley, C. M. Hogan, K. M. Johnson, E. Korotkikh, D. P. Lowry, D. Mandeno, R. M. McKay, J. A. Menking, T. R. Naish, C. Noerling, A. Ollive, A. Orsi, B. C. Proemse, A. R. Pyne, R. L. Pyne, J. Renwick, R. P. Scherer, S. Semper, M. Simonsen, S. B. Sneed, E. J. Steig, A. Tuohy, A. U. Venugopal, F. Valero-Delgado, J. Venkatesh, F. Wang, S. Wang, D. A. Winski, V. H. L. Winton, A. Whiteford, C. Xiao, J. Yang, X. Zhang, The Ross Sea Dipole – temperature, snow accumulation and sea ice variability in the Ross Sea region, Antarctica, over the past 2700 years. *Clim. Past* **14**, 193–214 (2018).

Acknowledgments: We acknowledge the High Performance Computer resources/Earth System Model Software/2024-EL-ZD-000191 from the National Key Scientific and Technological Infrastructure project “Earth System Numerical Simulation Facility” (EarthLab). **Funding:** K.M. and X.L. were supported by the National Science Fund for Distinguished Young Scholars (42325605) and the National Natural Science Foundation of China (42176243). C.W. was supported by the National Key Research & Development Program of China (2023YFF0805200) and the Science and Technology Innovation Project of Laoshan Laboratory (LSKJ20203300). D.M.H. was supported by the National Science Foundation grant PLR-2151295. N.Y. was supported by the Guangdong Basic and Applied Basic Research Foundation (2023B1515020084). **Author contributions:** Conceptualization: K.M., X.L., W.C., and J.L. Methodology: K.M., X.L., Y.W., L.G., and Y.H. Investigation: K.M. and X.L. Visualization: K.M., X.L., and Y.H. Supervision: X.L., W.C., and J.L. Writing—original draft: K.M. and X.L. Writing—review and editing: J.L., W.C., D.M.H., N.Y., Y.L., Y.W., L.G., G.S., and Y.H. **Competing interests:** The authors declare that they have no competing interests. **Data and materials availability:** Met Office Hadley Centre sea ice and SST (HadISST) dataset is available at www.metoffice.gov.uk/hadobs/hadisst/. The reconstructed snow accumulation datasets from (6) is available from <https://earth.gsfc.nasa.gov/cryo/data/antarctic-accumulation-reconstructions>. NOAA-CIRES-DOE 20th Century Reanalysis version 3 (20CRv3) is available at https://psl.noaa.gov/data/gridded/data.20thC_ReanV3.html. ECMWF's atmospheric reanalysis of the 20th century (ERA-20C) is available from <https://rda.ucar.edu/datasets/ds626.0/>. Further data analysis and figures are made with Matlab version R2018a (MathWorks Inc., 2020), accessible at <https://in.mathworks.com/products/matlab.html>. The ice core data and statistical analysis code are available at <https://zenodo.org/records/14015815>. All data needed to evaluate the conclusions in the paper are present in the paper and/or the Supplementary Materials.

Submitted 24 June 2024

Accepted 30 December 2024

Published 29 January 2025

10.1126/sciadv.adr2821